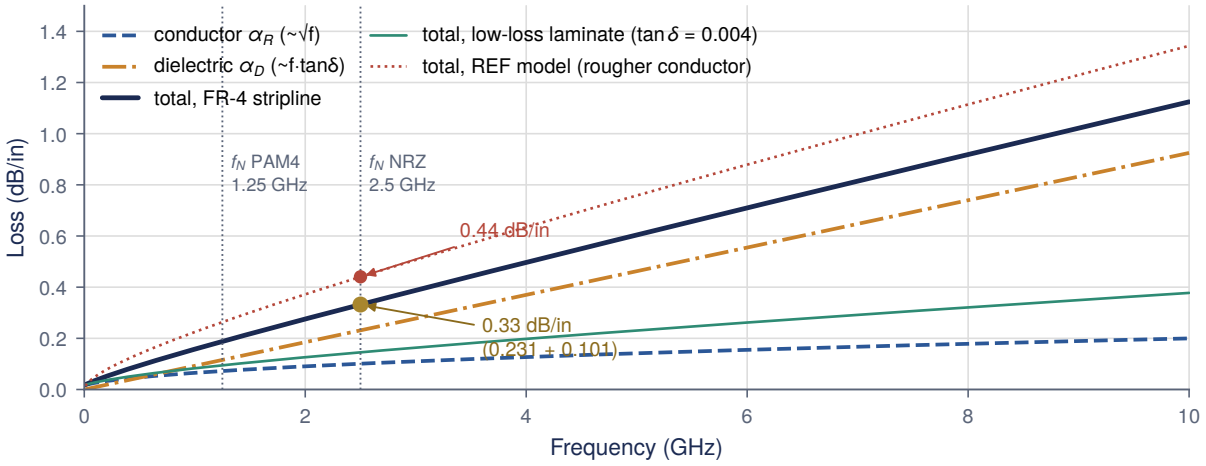
conductor α_R (~√f)
dielectric α_D (~f·tanδ)
total, FR-4 stripline
total, low-loss laminate (tan δ = 0.004)
total, REF model (rougher conductor)
f_N PAM4
1.25 GHz
f_N NRZ
2.5 GHz
0.44 dB/in
0.33 dB/in
(0.231 + 0.101)
Loss (dB/in)
Frequency (GHz)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
0
2
4
6
8
10